Rossmoyne Senior High School

Semester One Examination, 2020

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

In words ____________________

Circle your Teacher's Name: **ALVARO** **BESTALL** **FRASER-JONES** **GOH**

KOULIANOS **LUZUK** **RUDLAND** **TANDAY**

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Number of additional answer booklets used (if applicable):

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

	Number of questions available	Number of questions to be attempted	Working time (minutes)	Marks available	Percentage of exam
Section One Calculator—free	**9**	**9**	**50**	**54**	**36**
Section Two Calculator—assumed	15	15	100	96	64
				150	100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2020.* Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

   Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

   It is recommended that you **do not use pencil**, except in diagrams.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

5. The Formula Sheet is **not** handed in with your Question/Answer Booklet.

Section One: Calculator–free 36% (54 marks)

This section has **nine (9)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

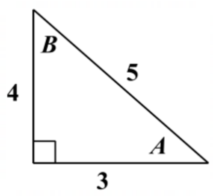
Working time: 50 minutes

Question 1 (6 marks)

(a) If $\sin 20° = p$, determine $\sin 340°$ in terms of p. (1 mark)

Solution
$\sin 340° = -p$
Specific behaviours
• ✓ correct answer only

(b) Given that $\angle A + \angle B = 90°$ and $\sin A = \frac{4}{5}$, evaluate $\cos(A - B)$. (3 marks)

Solution

$\cos A = 3/5 \quad \sin B = 3/5 \quad \cos B = 4/5$
$\cos(A-B) = \cos A\cos B + \sin A\sin B$
$= \frac{3}{5}.\frac{4}{5} + \frac{4}{5}.\frac{3}{5}$
$= \frac{24}{25}$
Specific behaviours
• ✓ determines correct values of cosA, sinB, cosB
• ✓ correct expansion of cos(A-B)
• ✓ substitutes correct values and correct final answer

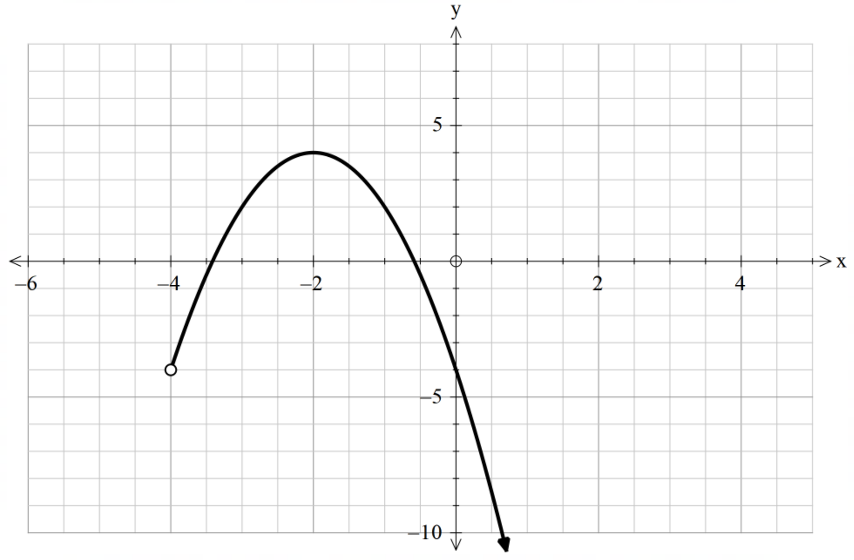
(c) Determine the exact value of the expression (2 marks)

$$2\tan\frac{\pi}{4} - \sin^2\frac{2\pi}{3}$$

Solution
$=2.1-\left[\frac{\sqrt{3}}{2}\right]^2$
$=2-\frac{3}{4}=\frac{5}{4}$
Specific behaviours
• ✓ correct sub ✓ correct answer

Question 2 (7 marks)

(a) State the **domain** of the function graphed below (2 marks)



Solution
$x > -4, \quad x \in R$.
Specific behaviours
✓ correct domain ✓ includes "Real"

(b) Solve $(x+1)^2 - 3 = x - 2$ (2 marks)

Solution
$x^2 + 2x + 1 - 3 = x - 2$ $x^2 + x = 0$ $x(x+1) = 0$ $x = 0, \qquad x = -1$
Specific behaviours
• ✓ expands and equates to zero • ✓ factorises and states both solutions

(c) A parabola has equation $y = -x^2 + 6x + 1$ (3 marks)
Rewrite the equation in the form

$$y = k(x-a)^2 + b$$

Solution
$y = -[x^2 - 6x + 9 - 9 - 1]$ $= -[(x-3)^2 - 10]$ $= -(x-3)^2 + 10$
Specific behaviours
• ✓ completing square • ✓ simplifying inside bracket • ✓ final answer

Or

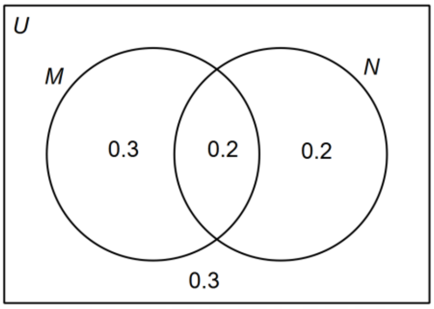
Solution
$y = -(x-3)^2 + 10$
Specific behaviours
• From formula sheet • ✓ $k = -1$ • ✓ $a = 2$ • ✓ $b = 10$

Question 3 (4 marks)

The following information is given:

- $P(M) = 0.5$
- $P(\overline{M \cup N}) = 0.3$
- $P(\overline{N}) = 0.6$

(a) Draw a Venn diagram to represent this information (2 marks)

Solution

Specific behaviours
• ✓ correct intersection 0.2 • ✓ correct 3 other values

(b) Hence, or otherwise determine

(i) $P(M \cup N)$ (1 mark)

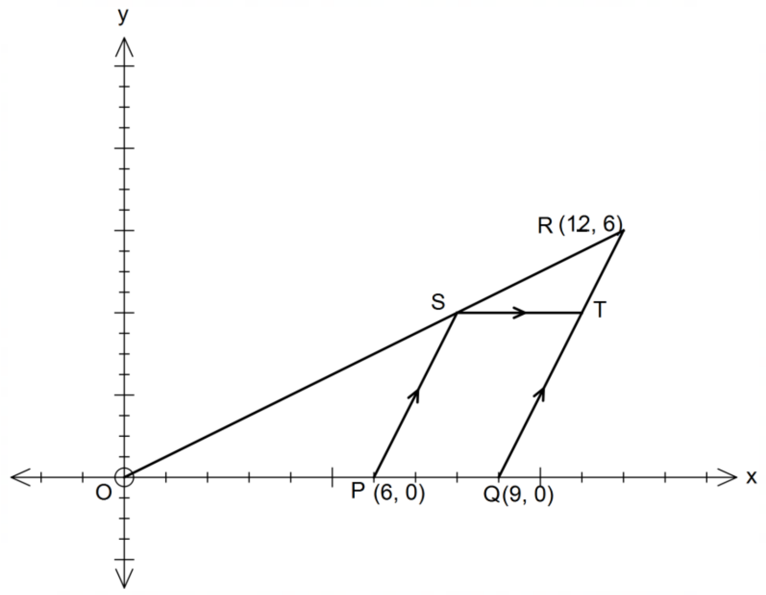
Solution
0.7
Specific behaviours
• ✓ correct answer

(ii) $P(M \mid N)$ (1 mark)

Solution
0.5
Specific behaviours
• ✓ correct answer

Question 4 (6 marks)

In the diagram P, Q and R are the points (6, 0), (9, 0) and (12, 6) respectively.
Line PS is parallel to QR. Line ST is parallel to the x-axis.
The equation of line OR is given by $x - 2y = 0$.



(a) Show that the equation of PS is $y = 2x - 12$. (3 marks)

Solution
From line QR: $m = \dfrac{6-0}{12-9} = 2$ Line PS: $y = 2x + c$ and $\therefore 0 = 2(6) + c$ $c = -12 \quad \therefore y = 2x - 12$
Specific behaviours
• ✓ correct gradient   • ✓ correct value of "c"   • ✓ correct equation

(b) Find the coordinates of point S. (3 marks)

Solution
$\dfrac{x}{2} = 2x - 12$ $-3x = -24$ $x = 8 \quad S(8, 4)$
Specific behaviours
• ✓ equating two lines   • ✓ solving for x   • ✓ state coordinates

Alt Solution
$S(8, 4)$
Specific behaviours
• ✓ states soln (from graph)   • ✓ confirms soln by subs into $y = x/2$ • ✓ confirms soln by subs into $y = 2x - 12$ If answer only 1/3

Question 5 (5 marks)

Daniel decides to plant trees as part of his #climateaction project. He has three jarrah trees and two paperbark trees. On each of the next three days, Monday, Tuesday and Wednesday, Daniel selects one tree at random to plant.

Find the probability that Daniel

(a) selects a jarrah tree to plant on Monday. (1 mark)

Solution
$P(J) = \frac{3}{5}$
Specific behaviours
• ✓ correct answer

(b) plants the same kind of tree on all three days. (1 mark)

Solution
Cannot plant 3 paperbarks $P(JJJ) = \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$
Specific behaviours
• ✓ correct answer

(c) does not plant the same kind of tree on consecutive days. (2 mark)

Solution
$P(JPJ \text{ or } PJP) = \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{3}{10}$
Specific behaviours
• ✓✓ correct answer

Solution
$P(JPJ) = \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} = \frac{1}{5}$ $\quad P(PJP) = \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{10}$
Specific behaviours
✓ correct answer for P(JPJ) **or** P(PJP)

(d) plants a Jarrah tree on Tuesday, given that he planted a Paperbark on Monday. (1 mark)

Solution
$P(J \mid P) = \frac{3}{4}$
Specific behaviours
• ✓ correct answer

Question 6 (5 marks)

Consider the graph with the equation $g(x) = -x^3 - 2x^2 + 7$.

(a) State the degree of the polynomial $-x^3 - 2x^2 + 7$. (1 mark)

Solution
Degree = 3 No mark for cubic

(b) Describe the behaviour of $g(x)$ when $x \to \infty$. (1 mark)

Solution
$g(x) \to -\infty$

Solution
• No mark for "decreasing" or "Becoming more negative"

(c) The graph of $y = g(x)$ is dilated horizontally by a factor of $\frac{1}{3}$ and then reflected through the $X-$ axis. The resulting graph is $y = h(x)$

(i) State $y = h(x)$ **(do not simplify)** (2 marks)

Solution	
$HD_{\times\frac{1}{3}}$	$= -(3x)^3 - 2(3x)^2 + 7$
R_x	$h(x) = (3x)^3 + 2(3x)^2 - 7$
Specific behaviours	
• ✓ correct substitution for HD	
• ✓ correct substitution for Reflection	
• Accept $h(x) = -[-(3x)^3 - 2(3x)^2 + 7]$	
• $h(x) = -g(3x)$ $1 mark$	

(ii) What is the $y-$ intercept of the graph of $y = h(x)$ (1 marks)

Solution
$h(0) = -7$ or $(0, -7)$

Question 7 **(6 marks)**

(a) Given that $x = 1$ is a solution to the equation $x^3 - 2x^2 - 5x + 6 = 0$, find the remaining solutions.

(3 marks)

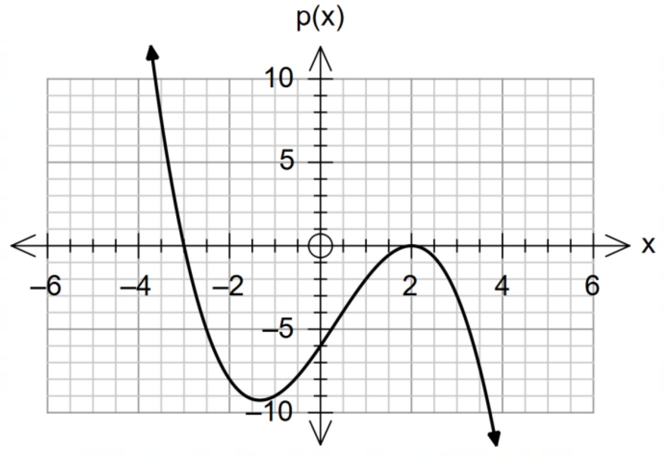
Solution
$x - 1\overline{)x^3 - 2x^2 - 5x + 6} = x^2 - x - 6$ $x^2 - x - 6 = (x + 2)(x - 3)$ $x = -2$ or $x = 3$
Specific behaviours
• ✓ correct quadratic   • ✓ factorising correctly   • ✓ correct solutions   Other methods (inspection, say) OK. Mark accordingly

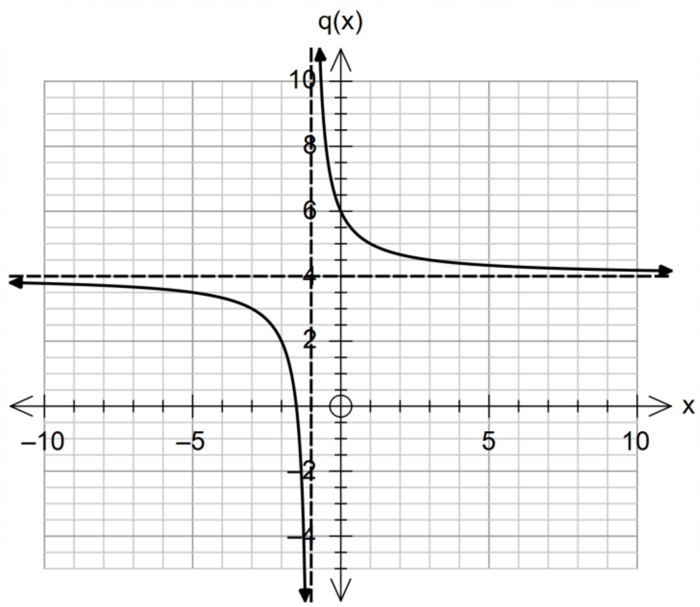
(b) Solve the equation $\sqrt{2}\sin(x - 20°) + 1 = 0$ where $0 \le x \le 360°$. (3 marks)

Solution
$\sin(x - 20°) = -\frac{1}{\sqrt{2}}$ Principle angle for $(x - 20°)$ $= \sin^{-1}(-\frac{1}{\sqrt{2}}) = -45°$ $x - 20° = 225°, 315°$ $x = 245°, 335°$
Specific behaviours
• ✓ determines an angle for $\pm\frac{1}{\sqrt{2}}$ that is , $45°$ *or* $-45°$ • ✓ determines one solution   • ✓ second solution   • $225°, 315°$ 2/3 marks

Question 8 (6 marks)

The graphs of $p(x) = a(x - b)^2(x - c)$ and $q(x) = d + \dfrac{e}{x - f}$ are shown below:





Find the values of *a, b, c, d, e* and *f*.

Solution

$p(x) = a(x-2)^2(x+3)$ $\quad\quad$ $q(x) = \dfrac{e}{x+1} + 4$

Sub (0,-6) $\quad\quad$ sub (0,6)

$\Rightarrow a = -\dfrac{1}{2}$ $\quad\quad$ $\Rightarrow$ e=2

$a = -\dfrac{1}{2}$ $\quad\quad$ $d = 4$

$b = 2$ $\quad\quad$ $f = -1$

$c = -3$ $\quad\quad$ $e = 2$

Specific behaviours

- ✓ one mark for each value
- Follow-through required

Question 9 (9 marks)

(a) A periodic function is defined by $f(x) = -2\sin(4x) + 2$

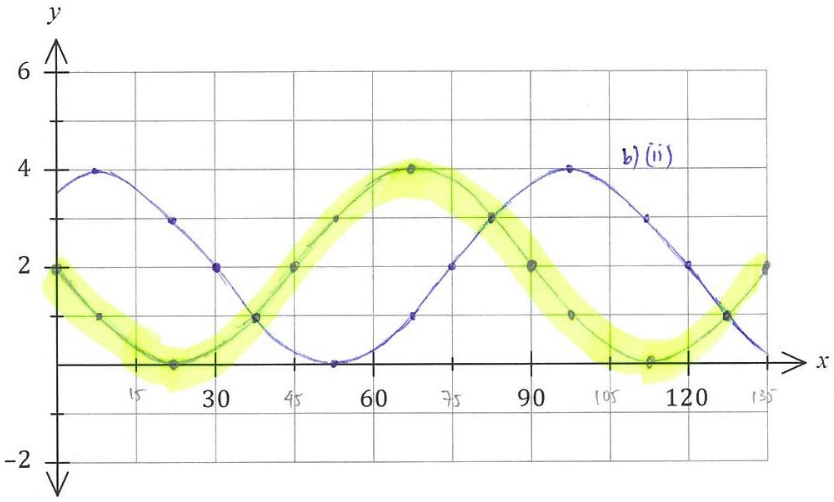
(i) State the amplitude of the function. (1 mark)

Solution
• Amplitude is 2.
Specific behaviours
• ✓ correct amplitude

(ii) State the period of the function in degrees. (1 mark)

Solution
• Period is $360 \div 4 = 90°$.
Specific behaviours
• ✓ correct period

(iii) Sketch the graph of $y = f(x)$ on the axes below. (4 marks)



Solution
• See graph (yellow)
Specific behaviours
• ✓ roots • ✓ locates maximum • ✓ smooth curve • ✓ endpoint Correct graph but incorrect vertical translation 2/4 Correct graph but incorrect period 2/4

(b) The graph of $f(x) = -2\sin(4x) + 2$ undergoes a horizontal translation of $30°$ units to the right.

(i) State the equation of the transformed graph. (1 marks)

Solution
• $f(x) = -2\sin\big(4(x - 30°)\big) + 2$ or $f(x) = -2\sin(4x - 120°) + 2$
Specific behaviours
• ✓ correct equation

(ii) Sketch this graph on the same set of axes above. (2 marks)

Solution
• See graph (blue)
Specific behaviours
• ✓ roots and maximum • ✓ endpoints and smooth curve

Additional working space

Question number(s): ……………………

Additional working space

Question number(s): ……………………